Passive-Aggressive Algorithms: Online Learning as a Per-Step Optimization

Yuval Lavie

August 10, 2026

Abstract

The perceptron corrects every mistake by the same fixed amount. The Passive-Aggressive family (Crammer, Dekel, Keshet, Shalev-Shwartz & Singer, 2006) instead *solves a tiny constrained optimization at every step*: stay as close as possible to the current weights while satisfying the incoming example's margin constraint. The KKT machinery of the optimization note delivers a closed-form step size, and two damped variants (PA-I, PA-II) buy noise robustness. Binary and multiclass versions, with the closed form verified against brute force and the noise-robustness race run in `pa.py`.

1 The idea: passive when satisfied, aggressive when not

Labels $y \in \{-1, +1\}$; margin target 1 via the hinge loss $\ell_t(w) = \max\big(0,\ 1 - y_t\, w^\top x_t\big)$. On each example solve

$$w_{t+1} \;=\; \arg\min_{w}\ \tfrac{1}{2}\|w - w_t\|^2 \quad \text{s.t.} \quad \ell_t(w) = 0. \tag{1}$$

Two regimes by construction. *Passive*: if $\ell_t(w_t) = 0$ the current weights are feasible and optimal — no update, unlike SGD, which nudges on every sample. *Aggressive*: otherwise, make the minimal change that fully satisfies the example. The learner is a sequence of projections onto half-spaces of "examples I now respect."

2 The closed form via KKT

Proposition 1. *When $\ell_t(w_t) > 0$, the solution of* (1) *is*

$$\boxed{w_{t+1} \;=\; w_t + \tau_t\, y_t\, x_t, \qquad \tau_t \;=\; \frac{\ell_t(w_t)}{\|x_t\|^2}.} \tag{2}$$

Proof. With the constraint active ($y_t w^\top x_t = 1$), the Lagrangian is $\frac{1}{2}\|w - w_t\|^2 + \tau\,(1 - y_t w^\top x_t)$ with multiplier $\tau \geq 0$. Stationarity in w: $w - w_t - \tau y_t x_t = 0$, so $w = w_t + \tau y_t x_t$ — the update direction is forced; only its length is free. Substitute into the active constraint: $y_t w_t^\top x_t + \tau\|x_t\|^2 = 1$, i.e. $\tau = (1 - y_t w_t^\top x_t)/\|x_t\|^2 = \ell_t(w_t)/\|x_t\|^2 > 0$, consistent with dual feasibility; complementary slackness holds since the constraint is met with equality. □

The perceptron's update is (2) with τ hard-coded to 1; PA sizes the correction by how badly the example is violated, normalized by its length — large enough to fix it exactly, never larger.

3 Noise: PA-I and PA-II

Aggressiveness is a liability under label noise: a mislabeled example forces (1) to satisfy a *wrong* constraint fully, however far that drags w (the script's race shows plain PA losing to the perceptron on a noisy stream). The fix is to let the constraint be violated at a price — add a slack variable $\xi \geq 0$, exactly the soft-margin move of the SVM note:

$$\text{PA-I:} \quad \min_{w,\xi} \ \tfrac{1}{2}\|w - w_t\|^2 + C\,\xi \ \text{ s.t. } \ell_t(w) \leq \xi \qquad \Longrightarrow \tau_t = \min\Big(C,\ \frac{\ell_t(w_t)}{\|x_t\|^2}\Big), \tag{3}$$

$$\text{PA-II:} \quad \min_{w,\xi} \ \tfrac{1}{2}\|w - w_t\|^2 + C\,\xi^2 \ \text{ s.t. } \ell_t(w) \leq \xi \qquad \Longrightarrow \tau_t = \frac{\ell_t(w_t)}{\|x_t\|^2 + \frac{1}{2C}}, \tag{4}$$

(derivations: same Lagrangian with the extra ξ term; the linear penalty caps the multiplier at C, the quadratic one shrinks it). C is the trust budget per example: $C \to \infty$ recovers plain PA; small C means no single example — honest or mislabeled — can move w far. In the script's race (7% flipped labels), PA-I with $C = 0.1$ recovers a clean-test error near the noise floor while plain PA thrashes.

4 Multiclass PA

Score with one vector per class; require the true class to beat the best wrong class by margin 1. With $\ell_t = \max\big(0,\, 1 - (w_{y_t} - w_{\hat{k}})^\top x_t\big)$ where $\hat{k}$ is the highest-scoring wrong class, the same derivation (the change now lives in two rows, whence the factor 2) gives

$$w_{y_t} \mathrel{+}= \tau_t x_t, \qquad w_{\hat{k}} \mathrel{-}= \tau_t x_t, \qquad \tau_t = \min\Big(C,\ \frac{\ell_t}{2\|x_t\|^2}\Big). \tag{5}$$

Promote the truth, demote the pretender — the multiclass perceptron's shape with an optimized step, and one more instance of the ladder this repository keeps climbing: softmax's soft $(p - e_y)$, the perceptron's hard ± 1, PA's "exactly as hard as needed."

Remark 1 (Guarantees)**.** *PA's analysis is regret-style, in the perceptron tradition: e.g. for separable streams PA enjoys a $(R/\gamma)^2$-type mistake bound, and PA-I/PA-II admit bounds that degrade gracefully with the cumulative hinge loss of the best fixed predictor in hindsight — competing with an oracle, not estimating a distribution. Distribution-free to the end.*